Vectors

Questions 1–28

2.45 — Question 1. It is given that A, B and C have the coordinates $(-2, 3, 5)$, $(4, -1, -3)$ and $(-1, 0, -8)$ respectively. Find

a) $\overrightarrow{AB}$
b) $\overrightarrow{BC}$
c) $\overrightarrow{AC}$
d) $\overrightarrow{BA}$

2.45 — Question 2. Find the magnitude of the following vectors:

a) $4\mathbf{i} + 5\mathbf{j} + 7\mathbf{k}$
b) $\begin{pmatrix} 6 \\ 7 \\ 1 \end{pmatrix}$
c) $5\mathbf{i} - 2\mathbf{j} + 12\mathbf{k}$
d) $\begin{pmatrix} a \\ b \\ 1 \end{pmatrix}$

2.45 — Question 3. Given each of the following position vectors, find the magnitude of the vector $\overrightarrow{AB}$.

a) $\overrightarrow{OA} = \begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 3 \\ 4 \\ -1 \end{pmatrix}$

b) $\overrightarrow{OA} = \mathbf{i} + 4\mathbf{j} - 5\mathbf{k}$ and $\overrightarrow{OB} = 3\mathbf{i} - \mathbf{j} + \mathbf{k}$

c) $\overrightarrow{OA} = \begin{pmatrix} 0 \\ 1 \\ 10 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 0 \\ -2 \\ 13 \end{pmatrix}$

2.45 — Question 4. Find the unit vector in the direction of the following vectors:

a) $2\mathbf{i} + 3\mathbf{j} - 7\mathbf{k}$

b) $\begin{pmatrix} 6 \\ -1 \\ 4 \end{pmatrix}$

c) $8\mathbf{i} - 6\mathbf{j} + 10\mathbf{k}$

d) $\frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$

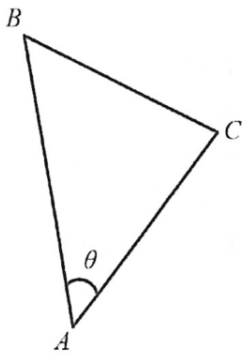
2.45 — Question 5. Given each of the following position vectors, find the unit vector in the direction of the vector $\overrightarrow{AB}$:

a) $\overrightarrow{OA} = \begin{pmatrix} 4 \\ 8 \\ -2 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 6 \\ 0 \\ -2 \end{pmatrix}$

b) $\overrightarrow{OA} = 2\mathbf{i} - 10\mathbf{j} - \mathbf{k}$ and $\overrightarrow{OB} = 7\mathbf{i} - 6\mathbf{j} + 2\mathbf{k}$

2.45 — Question 6. It is given that $\overrightarrow{AB} = \begin{pmatrix} -1 \\ 2 \\ 8 \end{pmatrix}$ and $\overrightarrow{BC} = \begin{pmatrix} 0 \\ 3 \\ -2 \end{pmatrix}$ and the angle between $\overrightarrow{AB}$ and $\overrightarrow{BC}$ is θ. Show that $\cos\theta = \frac{10}{\sqrt{897}}$.

2.45 — Question 7. The diagram shows the triangle ABC.



In each case, show that the value of the angle θ is the given value to the nearest degree:

a) $\theta = 18.3°$, $\overrightarrow{AB} = 2\mathbf{i} - 10\mathbf{j} + \mathbf{k}$ and $\overrightarrow{BC} = 7\mathbf{i} - 6\mathbf{j} + 2\mathbf{k}$

b) $\theta = 135.5°$, $\overrightarrow{AB} = \begin{pmatrix} 1 \\ 3 \\ -7 \end{pmatrix}$ and $\overrightarrow{AC} = \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}$

2.45 — Question 8. Position vectors of the points A, B and C are given as $\overrightarrow{OA} = 3\mathbf{i} - 2\mathbf{j} + \mathbf{k}$, $\overrightarrow{OB} = \mathbf{i} + 3\mathbf{j} + 5\mathbf{k}$ and $\overrightarrow{OC} = 8\mathbf{i} - 7\mathbf{j} - \mathbf{k}$. Given that $\overrightarrow{BC} = \overrightarrow{AD}$, find the position vector of the point D.

2.45 — Question 9. Three position vectors are given to be

$$\overrightarrow{OA} = \begin{pmatrix} 3 \\ 6 \\ -1 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 4 \\ 9 \\ 1 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OC} = \begin{pmatrix} 7 \\ 12 \\ p \end{pmatrix}$$

There is another point D and $\overrightarrow{AB} = \overrightarrow{BD}$.

a) Find the position vector of D.

b) Given $|\overrightarrow{AC}| = 9$, find the possible values of p.

2.45 — Question 10. Show that the coordinates $(7, 4, 0)$, $(12, 6, -1)$ and $(-13, -4, 4)$ are collinear.

2.45 — Question 11. Show that the coordinates $(6, 4, -3)$, $(8, 5, -3)$, $(12, 7, -15)$ are not collinear.

2.45 — Question 12. Three points A, B and C lie on a line. It is known that $\overrightarrow{OA} = \begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix}$ and $\overrightarrow{OB} = \begin{pmatrix} 5 \\ 12 \\ 2 \end{pmatrix}$ and furthermore that $|\overrightarrow{AC}| = 3|\overrightarrow{BC}|$. Find the possible position vectors of the point C.

2.45 — Question 13. The vector $\mathbf{a}$ is parallel to the given vector $\mathbf{p}$. In each case, find the vector $\mathbf{a}$:

a) $\mathbf{p} = 4\mathbf{i} - \mathbf{j} + 2\mathbf{k}$ and $|\mathbf{a}| = \sqrt{84}$
b) $\mathbf{p} = -\mathbf{i} + 7\mathbf{j} + 2\mathbf{k}$ and $|\mathbf{a}| = \sqrt{6}$
c) $\mathbf{p} = \mathbf{i} - 8\mathbf{j} - \mathbf{k}$ and $|\mathbf{a}| = 3\sqrt{66}$

2.45 — Question 14. Three position vectors are given as

$$\overrightarrow{OA} = \begin{pmatrix} 1 \\ 3 \\ -1 \end{pmatrix}, \quad \overrightarrow{OB} = \begin{pmatrix} 24 \\ -3 \\ 7 \end{pmatrix} \quad \text{and} \quad \overrightarrow{OC} = \begin{pmatrix} 5 \\ 1 \\ 2 \end{pmatrix}$$

Find

a) the midpoint of AB, m_{AB}
b) the midpoint of BC, m_{BC}
c) the distance between m_{AB} and m_{BC}
d) a unit vector in the direction of the line passing through m_{AB} and m_{BC}

The point D lies a distance of 1 unit away from A parallel to the line passing through the midpoints.

e) Find the possible coordinates of D.

2.45 — Question 15. For each of the following position vectors, state on which axes they lie:

a) $\begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$

b) $-12\mathbf{i}$

2.45 — Question 16. Points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$ where

$$\mathbf{a} = \begin{pmatrix} 1 \\ -4 \\ 2 \end{pmatrix} \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} 3 \\ 8 \\ -4 \end{pmatrix}$$

A third point C has position vector $k(\mathbf{a} + k\mathbf{b})$ where $k \neq 0$ and it is known that C lies on the x-axis.

a) Find the value of k.

b) Write down the position vector of C.

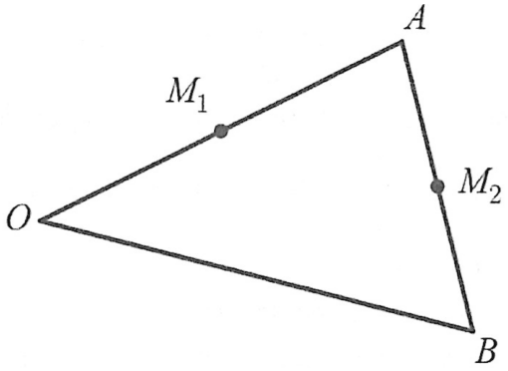
2.45 — Question 17. Points A and B have position vectors $\mathbf{a}$ and $\mathbf{b}$. A point C lies on AB such that $AC : CB$ is $6 : 1$. Show that the position vector of C is $\frac{1}{7}(\mathbf{a} + 6\mathbf{b})$.

2.45 — Question 18. Points A, B and C have position vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, where

$$\mathbf{a} = \begin{pmatrix} 3 \\ -6 \\ 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 5 \\ 4 \end{pmatrix} \quad \text{and} \quad \mathbf{c} = \begin{pmatrix} 14 \\ 7 \\ k \end{pmatrix}$$

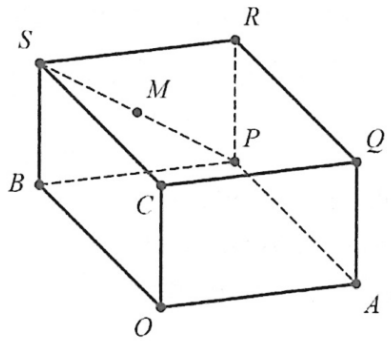
where k is a positive integer. It is known that $p\mathbf{a} + q\mathbf{b} = \mathbf{c}$ where p and q are positive integers. Find the values of p, q and k.

2.45 — Question 19. The diagram shows the triangle OAB with vertices labelled. The midpoint of $\overrightarrow{OA}$ is M_1 and the midpoint of $\overrightarrow{AB}$ is M_2.



By considering O as the origin, so that $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$, show that the line joining M_1 and M_2 is parallel to $\mathbf{b}$ and is half the length of $\mathbf{b}$.

2.45 — Question 20. The diagram shows the cuboid $OAPBCQRS$, with vertices labelled. The midpoint of $\overrightarrow{PS}$ is M.



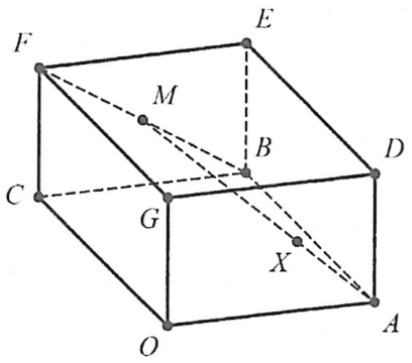
By considering O as the origin, so that $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$ and $\overrightarrow{OC} = \mathbf{c}$.

a) Show that $\overrightarrow{AM} = -\frac{1}{2}\mathbf{a} + \mathbf{b} + \frac{1}{2}\mathbf{c}$.

b) Find $\overrightarrow{PC}$ in terms of $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$.

c) Given that the lines $\overrightarrow{AM}$ and $\overrightarrow{PC}$ intersect, find the position vector of the point of intersection in terms of $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$.

2.45 — Question 20 (continued). Consider now, that the point X lies on AM such that $XM = 2AX$.



d) Find the position vector of X in terms of $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$.

2.45 — Question 21. Three points ABC form an isosceles triangle such that $|\overrightarrow{AB}| = |\overrightarrow{BC}|$, and it is known that the position vectors of A and B are

$$\mathbf{a} = \begin{pmatrix} 1 \\ 8 \\ 4 \end{pmatrix} \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} 1 \\ 5 \\ 7 \end{pmatrix}$$

respectively. The third point C has position vector $\mathbf{c} = \begin{pmatrix} 5 \\ 4 \\ k \end{pmatrix}$ where k is a positive constant to be found.

a) Find the possible values of k.
b) Find the exact area of ABC for both your values of k.

2.45 — Question 22. Four points A, B, C and D have position vectors

$$\mathbf{a} = 2\mathbf{i} + \mathbf{j} - 3\mathbf{k}, \quad \mathbf{b} = 6\mathbf{i} + 5\mathbf{k}, \quad \mathbf{c} = 5\mathbf{i} + 5\mathbf{j} + 7\mathbf{k} \quad \text{and} \quad \mathbf{d} = \mathbf{i} + 6\mathbf{j} - \mathbf{k}$$

Show that $ABCD$ is a parallelogram.

2.45 — Question 23. Three position vectors of points A, B and C are

$$\mathbf{a} = \begin{pmatrix} -1 \\ 5 \\ 8 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 4 \\ 13 \end{pmatrix} \quad \text{and} \quad \mathbf{c} = \begin{pmatrix} -2 \\ 5 \\ 19 \end{pmatrix}$$

There is another point D such that $ABCD$ forms a parallelogram and $\overrightarrow{AB} = \overrightarrow{DC}$.

a) Find the position vector of D.

b) Find the exact length of $\overrightarrow{AC}$.

2.45 — Question 24. Three position vectors of points A, B and C are

$$\mathbf{a} = 2\mathbf{i} - 4\mathbf{j} + 3\mathbf{k}, \quad \mathbf{b} = 6\mathbf{i} + 12\mathbf{j} + \mathbf{k} \quad \text{and} \quad \mathbf{c} = -\mathbf{i} + \mathbf{j} + 4\mathbf{k}$$

There is another point D such that $ABCD$ forms a parallelogram and $\overrightarrow{AB} = \overrightarrow{DC}$.

a) Find the position vector of D.

b) Find the length of $\overrightarrow{AB}$.

c) Find the length of $\overrightarrow{AC}$.

The angle between $\overrightarrow{AB}$ and $\overrightarrow{BC}$ is θ.

d) Show that $\cos\theta = \frac{105}{\sqrt{12351}}$.

e) Find the area of the parallelogram $ABCD$.

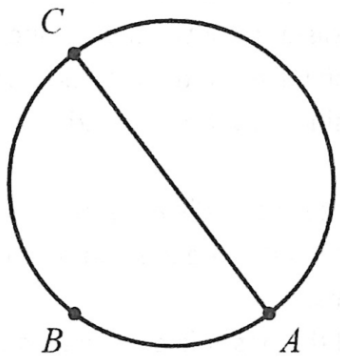
2.45 — Question 25. Four points A, B, C and D have position vectors

$$\mathbf{a} = \mathbf{i} - 4\mathbf{k}, \quad \mathbf{b} = 3\mathbf{i} + 3\mathbf{j} - 8\mathbf{k}, \quad \mathbf{c} = 8\mathbf{i} + 3\mathbf{j} - 10\mathbf{k} \quad \text{and} \quad \mathbf{d} = 6\mathbf{i} - 6\mathbf{k}$$

a) Show that $ABCD$ is a rhombus.

b) Find the area of the rhombus.

2.45 — Question 26. The position vectors of points A and B are given by $\mathbf{a} = \mathbf{i} + 5\mathbf{j} - 4\mathbf{k}$ and $\mathbf{b} = 7\mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$. The points A and B lie on a circle of radius $\sqrt{66}$. A further point $C(-1, -3, p)$ exists such that AC is the diameter of the circle.



a) Find the length of $\overrightarrow{BC}$.
b) Find the possible value(s) of p.

2.45 — Question 27. Four position vectors of points A, B, C and D are

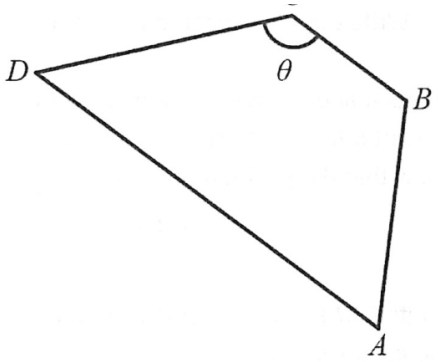
$$\mathbf{a} = -\mathbf{i} + 4\mathbf{j} + 7\mathbf{k}, \quad \mathbf{b} = \mathbf{i} + 5\mathbf{j} + 12\mathbf{k}, \quad \mathbf{c} = -2\mathbf{i} + 10\mathbf{j} + 13\mathbf{k} \quad \text{and} \quad \mathbf{d} = -7\mathbf{i} + 14\mathbf{j} + 9\mathbf{k}$$

Show that $ABCD$ is a trapezium.

2.45 — Question 28. Three position vectors of points A, B and C are

$$\mathbf{a} = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 8 \\ -3 \\ 3 \end{pmatrix} \quad \text{and} \quad \mathbf{c} = \begin{pmatrix} 9 \\ -1 \\ 7 \end{pmatrix}$$

There is an additional point D such that $ABCD$ is a trapezium where $\overrightarrow{BC} = \frac{1}{3}\overrightarrow{AD}$.



a) Find the position vector of D.

The angle between BC and CD, θ, is given by $\cos\theta = -\frac{31}{7\sqrt{30}}$.

b) Show that the exact area of the trapezium $ABCD$ is $2\sqrt{509}$ units^2.